

Name: _____

Date: _____

TRIGONOMETRY, (SOH CAH TOA — FINDING ANGLES)

LO: I can use inverse trigonometric functions to find missing angles in right-angled triangles.

Finding angles

To find a missing **angle** in a right-angled triangle, use the **inverse trig functions**: $\sin^{-1}$, $\cos^{-1}$, or $\tan^{-1}$.

SOH CAH TOA: Choose the ratio that uses the two known sides. Then apply the inverse function to find the angle.

Quick examples

●	$\sin x = \frac{5}{10}$ $x = \sin^{-1}(0.5) = 30^\circ$	inverse sin
●	$\cos x = \frac{8}{10}$ $x = \cos^{-1}(0.8) = 36.9^\circ$	inverse cos
●	$\tan x = \frac{7}{4}$ $x = \tan^{-1}(1.75) = 60.3^\circ$	inverse tan
●	$\sin x = \frac{5}{10}$ means $x = 0.5^\circ$	must use inverse sin

Key idea

Identify which sides you know (O, A, H), choose the correct ratio, then use the inverse function.

$$\theta = \sin^{-1}\left(\frac{O}{H}\right)$$

SOH: $\sin = \text{Opp/Hyp}$, CAH: $\cos = \text{Adj/Hyp}$, TOA: $\tan = \text{Opp/Adj}$

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - using sin

Find angle x. Opposite = 6 cm, hypotenuse = 12 cm.

Opp and Hyp: use sin

$$\sin x = \frac{6}{12} = 0.5$$

$$x = \sin^{-1}(0.5) = 30^\circ$$

$$x = 30^\circ$$

Label the sides relative to the angle, then choose the correct ratio.

Example 2 - using cos

Find angle x. Adjacent = 8 cm, hypotenuse = 10 cm.

Adj and Hyp: use cos

$$\cos x = \frac{8}{10} = 0.8$$

$$x = \cos^{-1}(0.8) = 36.9^\circ$$

$$x = 36.9^\circ$$

CAH: $\cos = \text{adjacent} / \text{hypotenuse}$.

Example 3 - using tan

Find angle x. Opposite = 9 cm, adjacent = 5 cm.

Opp and Adj: use tan

$$\tan x = \frac{9}{5} = 1.8$$

$$x = \tan^{-1}(1.8) = 60.9^\circ$$

$$x = 60.9^\circ$$

TOA: $\tan = \text{opposite} / \text{adjacent}$.

Example 4 - third angle

In a right triangle, one angle is 53.1° . Find the other acute angle.

Angles sum to 180°

$$\text{Third} = 180 - 90 - 53.1$$

$$= 36.9^\circ$$

$$36.9^\circ$$

Angles in a triangle sum to 180° .

YOUR TURN – PRACTICE (deliberate practice)

1. Using sin inverse

Find angle x (1 dp). Given opp and hyp.

- a) opp = 5, hyp = 10
- b) opp = 7, hyp = 14
- c) opp = 4, hyp = 9
- d) opp = 12, hyp = 15

2. Using cos inverse

Find angle x (1 dp). Given adj and hyp.

- a) adj = 6, hyp = 10
- b) adj = 9, hyp = 12
- c) adj = 3, hyp = 7
- d) adj = 5, hyp = 13

3. Using tan inverse

Find angle x (1 dp). Given opp and adj.

- a) opp = 5, adj = 5
- b) opp = 8, adj = 3
- c) opp = 4, adj = 7
- d) opp = 11, adj = 6

4. Choose the ratio

Find angle x (1 dp).

- a) opp = 6, hyp = 9
- b) adj = 5, hyp = 13
- c) opp = 10, adj = 4
- d) adj = 7, opp = 7

5. In context

Find each angle to 1 dp.

a) Ladder 5 m, reaches 4 m up wall. Angle with ground?

b) Ramp rises 2 m over 8 m horizontal. Angle?

c) Kite string 30 m, height 18 m. Angle of elevation?

d) Shadow 12 m, pole 5 m. Sun angle?

e) Roof height 3 m, half-span 5 m. Pitch angle?

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) To find an angle, use the inverse trig function ☐

Correct answer: _____

Reason: _____

b) $\sin^{-1}(0.5) = 2$ ☐

Correct answer: _____

Reason: _____

c) tan is used when you know opposite and adjacent ☐

Correct answer: _____

Reason: _____

d) $\cos^{-1}$ is used when you know opp and hyp ☐

Correct answer: _____

Reason: _____

e) Angles in a triangle sum to 180° ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A rectangular field is **80 m by 60 m**. Find the angle between the diagonal and the longer side.

Expression: _____

Simplified: _____

b) From the top of a **25 m** cliff, the angle of depression to a boat is **32°** . How far is the boat from the base of the cliff?

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

